Brachydactyly-Syndactyly Syndrome (MONDO:0012544): Comprehensive Disease Characteristics Report

Summary

Brachydactyly-syndactyly syndrome (OMIM 610713; MONDO:0012544) is a rare, congenital, autosomal dominant limb-malformation syndrome caused by heterozygous mutations in **HOXD13**, the homeodomain transcription factor on chromosome 2q31.1 that patterns the distal limb (autopod). The disorder combines short digits (**brachydactyly**, overlapping types A4, D, and E) with digit fusion (**syndactyly**, with mild syndactyly of toes 2–3), and was formally defined by Zhao et al. in 2007 through the study of two large Han Chinese families — one with syndactyly type V and the other with the overlapping brachydactyly-syndactyly phenotype (PMID: 17236141). The syndrome sits on a continuous *HOXD13* allelic spectrum that also encompasses synpolydactyly type 1 (SPD1, OMIM 186000) and syndactyly type V (SDTY5, OMIM 186300).

The central mechanistic insight of this investigation is that **different classes of HOXD13 mutation act through distinct molecular mechanisms that converge on dysregulated autopod chondrogenesis**. Polyalanine-tract expansions in the N-terminal intrinsically disordered region act through a dominant-negative / gain-of-function mechanism now understood to involve altered **transcriptional condensate phase separation** ("unblending"), while truncating, frameshift, splice-site, and DNA-binding-domain missense variants act through **haploinsufficiency / loss of function**. These converge downstream onto a shared set of limb-patterning pathways: the EPHA7 receptor tyrosine kinase (a direct HOXD13 transcriptional target), retinoic-acid synthesis (RALDH2/Raldh2), WNT5A-planar-cell-polarity signaling controlling chondrocyte polarity, and the SHH/GREM1/FGF/BMP2 feedback network — all validated in *Spdh*, *Dyc*, and *Hoxd13*-Q50R mouse models.

Clinically, the syndrome shows **incomplete penetrance** and **highly variable, often asymmetric expressivity**, with phenotype severity correlating positively with alanine-repeat length. Lifespan is normal, morbidity is confined to hand and foot function and cosmesis, and management is entirely **surgical and supportive** (syndactyly release, polydactyly/

brachydactyly correction, hand/occupational therapy) plus genetic counseling — there is no disease-modifying pharmacotherapy. Diagnosis is clinical/radiographic, confirmed by HOXD13 molecular testing including triplet-repeat-length analysis, and prenatal diagnosis is feasible by combined ultrasound and mutation analysis.

Section 1: Disease Information

Overview. Brachydactyly-syndactyly syndrome is a Mendelian, congenital limb-malformation disorder of the distal limb (hands and feet). It is characterized by shortening of digits (brachydactyly) combined with incomplete separation/fusion of digits (syndactyly). It was defined as a novel clinical entity by Zhao et al. (2007) in a Han Chinese family whose limb features overlapped brachydactyly types A4, D, and E together with mild syndactyly of toes 2 and 3 (PMID: 17236141): *"We describe two large Han Chinese families with different limb malformations, one with syndactyly type V and the other with limb features overlapping brachydactyly types A4, D, and E and mild syndactyly of toes 2 and 3."*

Key identifiers.

Resource	Identifier
MONDO	MONDO:0012544
OMIM	610713 (brachydactyly-syndactyly syndrome)
Related OMIM phenotypes	186000 (Synpolydactyly type 1, SPD1); 186300 (Syndactyly type V)
Gene	HOXD13 (HGNC:5136; OMIM *142989); chromosome 2q31.1
ICD-10	Q70–Q74 range (congenital malformations of limbs); Q70 (syndactyly)
MeSH	Synpolydactyly / Syndactyly / Brachydactyly

Synonyms / alternative names. Brachydactyly-syndactyly syndrome; HOXD13-related limb malformation; the broader allelic series is often referred to as the **HOXD13-associated synpolydactyly (SPD1) spectrum**. Related named allelic entities include syndactyly type V (SDTY5) and synpolydactyly type 1.

Source of information. The information is derived from **aggregated disease-level resources** (OMIM, Orphanet, primary literature) and detailed **individual-patient / family-level clinical-genetic reports** — notably large multigenerational pedigrees (Han Chinese, Pakistani, Polish, and others) and the recent 87-family cohort study of Gottschalk et al. (2023, PMID: 37427568), which pooled 38 new and 49 published families.

Section 2: Etiology

Primary cause — genetic. The disorder is a monogenic, autosomal dominant condition caused by **heterozygous mutations in HOXD13**. There is no known environmental, infectious, or toxic cause; it is a developmental (morphogenetic) disorder of embryonic limb patterning.

Genetic risk factors (causal variants). Causal lesions in HOXD13 fall into several classes:

- **Polyalanine-tract expansions** in the imperfect GCN-triplet-containing exon 1 (most common; e.g., +7 to +9 alanine additions) — the predominant cause of the synpolydactyly end of the spectrum.
- **Polyalanine contraction** — a 21-bp deletion producing a 7-residue polyalanine contraction defined the original brachydactyly-syndactyly family (PMID: 17236141): *"we detected a deletion of 21 bp in the imperfect GCN...triplet-containing exon 1 of HOXD13, which results in a polyalanine contraction of seven residues."*
- **Homeodomain missense variants** (e.g., p.Q317R, p.I47L, p.Q50R, p.Thr313Arg) that abolish DNA binding and transactivation.
- **Truncating variants** — nonsense (p.Gln248Ter, p.Gln287Ter), frameshift (p.Gly237fs), and splice-donor-site mutations acting through haploinsufficiency.

Modifier genes and susceptibility loci. Beyond HOXD13 itself, other genes can produce overlapping phenotypes: **PTHLH** (brachydactyly type E, PMID: 31283647, PMID: 26763883), **FBLN1** (rare SPD deletion), **TTC30B** (candidate SPD variant, PMID: 31306809), and in broader congenital limb-anomaly cohorts **LMBR1, GJA1, TP63** (PMID: 39639541). GLI3 (GLI3R) interacts with HOXD13 in limb pre-patterning and modulates the phenotype (PMID: 22373878).

Environmental / protective factors / gene-environment interactions. None are established for this Mendelian disorder. No environmental toxins, lifestyle factors, maternal exposures, or protective alleles are implicated. The one notable experimental "environmental" modifier is

intrauterine retinoic acid supplementation, which rescued the polydactyly phenotype in the mouse *spdh* model (PMID: 19075394) — a laboratory finding, not a human exposure. **This category is largely not applicable** to a congenital Mendelian limb-patterning syndrome.

Section 3: Phenotypes

The core phenotype is a combination of **physical/structural congenital malformations** of the autopod. All are **congenital (present at birth)**, **stable/non-progressive**, and affect bones and soft tissues of the hands and feet.

Phenotype	HPO term (suggested)	Type	Onset	Severity	Frequency
Brachydactyly (short digits)	HP:0001156	Physical/ skeletal	Congenital	Mild–severe, variable	Core feature
Short metacarpal	HP:0010049	Physical/ skeletal	Congenital	Variable	Common
Short phalanx of finger	HP:0009803	Physical/ skeletal	Congenital	Variable	Common
Cutaneous syndactyly	HP:0010554	Physical/ soft tissue	Congenital	Mild (toes 2–3) to complex	Core feature
Finger syndactyly	HP:0006101	Physical	Congenital	Variable	Common
Toe syndactyly	HP:0001770	Physical	Congenital	Mild	Core feature
Synpolydactyly	HP:0009855 (related)	Physical	Congenital	Variable	Spectrum end
Clinodactyly	HP:0030084	Physical	Congenital	Mild	Reported
Metacarpal-to-carpal transformation	—	Physical/ skeletal	Congenital	Severe	Homozygotes

Phenotype characteristics. Onset is uniformly **congenital/neonatal**; the malformations are **non-progressive and stable** across life. Severity ranges from **mild** (short 5th metacarpal, clinodactyly, subtle toe webbing) to **severe** (osseous synpolydactyly, metacarpal-to-carpal transformation in homozygotes). Gottschalk et al. documented the full breadth (PMID: 37427568): *"Phenotypes ranged from unaffected heterozygotes to severe osseous synpolydactyly, with intra- and inter-familial heterogeneity and asymmetry."* A pathognomonic feature of homozygous SPD1 is a relatively long, medially deviated big toe with/without cupping of the forefoot (PMID: 32509852).

Quality-of-life impact. Impact is confined to **hand and foot function and appearance** — manual dexterity, grip, fine motor development, footwear fit, and psychosocial/cosmetic concerns. There is no systemic, cognitive, or life-limiting impact in the non-syndromic HOXD13 form. (Syndromic brachydactyly+syndactyly phenotypes with intellectual disability, microcephaly, and short stature — e.g., PMID: 21416592 — are *distinct* entities and should not be conflated with HOXD13-related disease.)

Section 4: Genetic / Molecular Information

Causal gene. HOXD13 (Homeobox D13; HGNC:5136; OMIM *142989), the most 5′ gene of the HOXD cluster on chromosome 2q31.1. HOXD13 encodes a homeodomain transcription factor that, with HOXA13, is specifically expressed in the developing autopod (PMID: 8898214):* "Only two genes from homology group 13 (Hoxa-13 and Hoxd-13) are specifically expressed in the developing distal region (the autopod), which displays the most complex and evolutionarily flexible pattern among limb 'segments'."*

Pathogenic variant classes and functional consequences.

Variant (example)	Type	Mechanism	Phenotype	PMID
21-bp deletion → polyAla −7	In-frame deletion	Altered condensation	Brachydactyly-syndactyly	17236141
polyAla expansion (+7 to +9 Ala)	In-frame expansion	Dominant-negative / phase separation	Synpolydactyly	37427568, 32386547
c.184_210dup (+9 Ala)	Expansion	Dominant-negative	SPD1	22161087
c.950A>G (p.Q317R)	Homeodomain missense	Loss of DNA binding; abolishes EPHA7 transactivation	Syndactyly type V	17236141
p.I47L	Homeodomain missense	Loss of DNA binding / EphA7 transactivation	Brachydactyly-polydactyly	16314414
c.938C>G (p.Thr313Arg), homozygous	DNA-binding missense	Abolishes DNA binding (EMSA); non-penetrant in heterozygotes	Severe brachydactyly, metacarpal-to-carpal transformation	26581570
c.708_720del13 (p.Gly237fs)	Frameshift	Premature termination / LOF	BDA3 + atypical BDA4	32789964
c.859C>T (p.Gln287Ter)	Nonsense	Haploinsufficiency via NMD	SPD (mild)	31611522
c.781+1G>A (p.G190fsX4)	Splice-donor	Truncated non-DNA-binding protein / LOF	SPD + cortical bone thinning	24055421
p.G11A	N-terminal missense	Destabilizes GLI3R; alters protein half-life	SPD	22373878

Variant classification (ACMG/AMP). Recurrent polyalanine expansions and the homeodomain missense variants are classified **pathogenic**; truncating variants causing haploinsufficiency are **pathogenic/likely pathogenic**. Interpretation is complicated by incomplete penetrance.

Allele frequency / origin. Pathogenic HOXD13 variants are **germline**, private or recurrent within families, and essentially absent from population databases (gnomAD) given their severe developmental consequence. There is **no somatic contribution**. The polyalanine expansions arise by replication-based mechanisms — both **unequal crossover** (duplication; *spdh*) and **fork stalling and template switching (FoSTeS)** (*Dyc* model, PMID: 19546318).

Functional consequences. Two divergent mechanisms (see Section 6): **dominant-negative/gain-of-function** (polyalanine expansions) versus **loss-of-function/haploinsufficiency** (truncating and DNA-binding missense). As summarized by Zhang et al., *"missense and out-of-frame deletion mutations in the HOXD13 gene are also known to cause SPD, and the mechanism responsible for the phenotype appears to be haploinsufficiency"* (PMID: 21814222).

Modifier genes, epigenetics, chromosomal abnormalities. Modifiers include GLI3 (GLI3R), HOXA13 (partially redundant paralog), and PTHLH. Epigenetic contributions are not a primary feature. Large interstitial deletions at distinct loci (e.g., 12p11.22-12.1 encompassing PTHLH) can cause phenocopy brachydactyly type E (PMID: 31283647).

Section 5: Environmental Information

Not applicable as a cause. This is a Mendelian developmental disorder with no established environmental, lifestyle, or infectious etiology. No toxins, radiation, pollution, occupational exposures, dietary factors, or pathogens are implicated in causation or triggering. The sole experimental environmental modifier is **retinoic acid**, demonstrated only as an *in utero* pharmacological rescue in the mouse *spdh* model (PMID: 19075394) — mechanistically informative but not a human environmental risk/protective factor.

Section 6: Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation)

1. A **heterozygous HOXD13 mutation** (polyalanine expansion/contraction in exon 1, OR a homeodomain/truncating loss-of-function variant) is present in the germline and expressed in autopod mesenchyme. *(demonstrated)*

2. **Branch A — polyalanine expansion:** The altered N-terminal intrinsically disordered region **changes the phase-separation / co-condensation behavior** of HOXD13, perturbing the composition of HOXD13-containing transcriptional condensates and **interfering with other HOXD proteins** (dominant-negative). *(demonstrated in vitro and in mouse; PMID: 32386547, PMID: 11543619)*

Branch B — truncating / DNA-binding missense: The variant produces a **non-DNA-binding or NMD-degraded protein**, reducing functional HOXD13 dosage → **haploinsufficiency**. *(demonstrated; PMID: 21814222, PMID: 24055421)*

1. Both branches **dysregulate the HOXD13 transcriptional program** in digit-forming mesenchyme. *(demonstrated)*
2. This **leads to loss of direct activation of EPHA7**, a receptor tyrosine kinase target normally transactivated by HOXD13/HOXA13 at paralog-13 Hox binding sites. *(demonstrated; PMID: 16314414)*
3. In parallel, mutant HOXD13 **reduces RALDH2-dependent retinoic-acid synthesis** and promotes **ectopic interdigital chondrogenesis** (increased Sox9), **resulting in** extra/fused digits. *(demonstrated in spdh mouse; PMID: 19075394)*
4. Mutant HOXD13 also **fails to induce WNT5A/WNT5B**, **leading to** randomized chondrocyte polarity and mislocalized planar-cell-polarity components (DVL2, PRICKLE1) with perichondrial WNT/β-catenin upregulation → **metacarpal-to-carpal transformation** and loss of cortical bone/perichondrium. *(demonstrated in spdh mouse; PMID: 24161848)*
5. For homeodomain mutants, **disruption of the SHH/GREM1/AER-FGF8 feedback loop with BMP2 downregulation** occurs in the limb bud, **resulting in** combined syndactyly, brachydactyly, and polydactyly. *(demonstrated in Hoxd13-Q50R mouse; PMID: 38561387)*
6. Collectively, defective chondrocyte differentiation, reduced proliferation, abnormal joint patterning, and disrupted digit separation **produce the congenital brachydactyly-syndactyly phenotype**. *(demonstrated)*

Molecular pathways and detail

- **Transcriptional condensate phase separation (upstream, dominant-negative branch).** Basu et al. (Cell 2020) showed that alanine-repeat expansions in the HOXD13 intrinsically disordered region *"alter its phase separation capacity and its capacity to co-condense with transcriptional co-activators"* and that *"HOXD13 repeat expansions perturb the composition of HOXD13-containing condensates in vitro and in vivo and alter the transcriptional program*

in a cell-specific manner in a mouse model of synpolydactyly" (PMID: 32386547). GO terms: GO:0006355 (regulation of DNA-templated transcription), GO:0032991 (protein-containing complex).

- **EPHA7 signaling (direct target).** Salsi & Zappavigna demonstrated *"HOXD13 and HOXA13 activate transcription from the EphA7 promoter"* and that the pathogenic I47L variant *"does not bind to in vivo, and fails to transactivate the EphA7 promoter"* (PMID: 16314414). EphA7 governs differential cell affinity/sorting in the autopod.
- **SHH / ZRS / GREM1 / FGF / BMP2 (AP and feedback patterning).** HOXD13 cooperates with Hand2 at the ZRS enhancer to activate Shh (PMID: 20386744): *"Hand2 is part of the protein complexes containing Hoxd13, another essential regulator of Shh activation in limb buds."* Hoxd13-Q50R mice showed *"ectopically expressed Shh, Grem1 and Fgf8 and down-regulated Bmp2 in the embryonic limb bud at E10.5 to E12.5"* (PMID: 38561387).
- **Retinoic-acid pathway (downstream, chondrogenesis).** *"mutated Hoxd13 causes polydactyly in SPD by inducing extraneous interdigital chondrogenesis, both directly and indirectly, via a reduction in RA levels"* (PMID: 19075394). CHEBI: retinoic acid (CHEBI:26536).
- **WNT5A / planar cell polarity (chondrocyte orientation).** *"Wnt5a and the closely related Wnt5b were downregulated in spdh handplates, and HOXD13 induced expression of both genes in vitro"* and *"mislocalization of core planar cell polarity (PCP) components DVL2 and PRICKLE1 in spdh metacarpals indicating a defect in the WNT/PCP pathway"* (PMID: 24161848).
- **GLI3R interaction.** The N-terminal p.G11A variant destabilizes GLI3 repressor during limb pre-patterning (PMID: 22373878).

Cellular processes: chondrocyte differentiation and proliferation (reduced; PMID: 11850178), chondrocyte planar polarity, interdigital apoptosis (disrupted separation), cell sorting/ differential affinity, and joint patterning. **Suggested GO terms:** GO:0048704 (embryonic skeletal system morphogenesis), GO:0035108 (limb morphogenesis), GO:0002063 (chondrocyte development), GO:0001501 (skeletal system development), GO:0060173 (limb development). **Suggested CL terms:** CL:0000138 (chondrocyte), CL:0000134 (mesenchymal stem cell), CL:0000057 (fibroblast / limb mesenchymal cell).

Upstream vs downstream. Upstream: the HOXD13 mutation and altered transcriptional output. Midstream: EPHA7, SHH/GREM1/FGF/BMP2, WNT5A-PCP, and RA pathways. Downstream: chondrocyte polarity/differentiation/proliferation defects, altered digit separation and condensation → gross skeletal malformation.

Section 7: Anatomical Structures Affected

Organ/system level. The disorder affects the **musculoskeletal system**, specifically the **distal limbs (autopod)** — hands and feet. There is no primary involvement of cardiovascular, nervous, respiratory, digestive, or endocrine systems in the non-syndromic HOXD13 form.

- **Primary structures:** phalanges, metacarpals, metatarsals, carpal/tarsal bones, interdigital webbing (skin/soft tissue), phalangeal joints.
- **UBERON terms:** UBERON:0002398 (manus/hand), UBERON:0002387 (pes/foot), UBERON:0002544 (digit), UBERON:0001436 (metacarpal bone), UBERON:0003221 (phalanx), UBERON:0006048 (autopod region).

Tissue and cell level. Cartilage and bone tissue (endochondral skeleton) and **interdigital mesenchyme/skin** are affected. Key cell populations: **chondrocytes** (CL:0000138) in growth plates and perichondrium, **limb-bud mesenchymal cells** (CL:0000134), and perichondrial cells. Metacarpal-to-carpal transformation reflects a tissue-identity switch in cartilage elements (PMID: 24161848, PMID: 26581570). Cortical bone thinning is seen with certain splice variants (PMID: 24055421).

Subcellular level. The primary subcellular locus is the **nucleus** (GO:0005634), where HOXD13 functions as a transcription factor, and the associated **biomolecular condensates / nuclear bodies** (transcriptional condensates). GO cellular-component terms: GO:0005634 (nucleus), GO:0000785 (chromatin), GO:0032991 (protein-containing complex).

Localization and lateralization. Both **hands and feet** are affected, typically **bilateral** but frequently **asymmetric** between the two sides and between individuals (PMID: 37427568). The feet (especially toes 2–3 webbing and the big-toe anomaly) and the fourth web space / 4th–5th digits are characteristically involved.

Section 8: Temporal Development

Onset. Congenital — the malformations are established during embryonic limb patterning (approximately E10.5–E12.5 in mouse models; first trimester in humans) and are present at birth. Onset pattern is **developmental/static**, not acute or progressive.

Progression. The structural anomalies are **stable and non-progressive** across life; they do not worsen biologically, though functional/cosmetic consequences persist lifelong unless surgically corrected. There are **no disease stages** in the oncologic or neurodegenerative sense. Growth-plate and chondrocyte abnormalities in the *spdh* mouse persist postnatally and correlate with continued mutant Hoxd13 expression (PMID: 11850178).

Patterns and critical periods. There is no remission. The **critical period of vulnerability** is embryonic autopod patterning. The **critical window of therapeutic opportunity** in experimental models is in utero (retinoic-acid rescue in mice, PMID: 19075394); in humans, the practical intervention window is postnatal surgical correction, typically in infancy/early childhood.

Section 9: Inheritance and Population

Epidemiology. This is a **rare disorder**. Synpolydactyly (the broader HOXD13 spectrum) has an estimated prevalence on the order of ~1 in 300,000 (Orphanet-level estimates); the specific brachydactyly-syndactyly syndrome is defined from individual large pedigrees and precise population prevalence/incidence figures are **not established**. Hand syndactyly overall (all causes) is the most common upper-extremity congenital malformation (PMID: 40206497).

Inheritance pattern. Autosomal dominant (HOXD13 at 2q31.1). Severe forms (e.g., metacarpal-to-carpal transformation) can appear in **homozygotes**, with the heterozygous state non-penetrant (PMID: 26581570).

Penetrance and expressivity. Characterized by **incomplete penetrance** and **highly variable expressivity**, including asymmetry and unaffected mutation carriers (PMID: 37427568, PMID: 21814222). Severity correlates with **alanine-repeat length** (PMID: 37427568): *"Computer-aided analysis only corroborated a positive correlation between alanine repeat length and phenotype severity."*

Genetic anticipation, mosaicism, founder effects. No genetic anticipation is established (polyalanine tracts are relatively stable compared with polyglutamine repeat disorders). Germline mosaicism is theoretically possible but not prominently documented. No strong founder effects; recurrent expansions arise independently via unequal crossover or FoSTeS (PMID: 19546318). **Consanguinity** is relevant for the rare homozygous severe forms.

Population demographics. Families reported across diverse ancestries — Han Chinese (PMID: 17236141, PMID: 22161087), Pakistani (PMID: 39639541), Polish (PMID: 26763883), and the pan-ancestry Gottschalk cohort. No strong geographic clustering. **Sex ratio** is approximately equal (autosomal). **Age distribution:** congenital, affecting all ages equally since the phenotype is lifelong.

Section 10: Diagnostics

Clinical/radiographic diagnosis. Diagnosis begins with **clinical examination** (short and/or fused digits, extra digits) and **hand/foot radiographs** demonstrating short or fused metacarpals/metatarsals, abnormal/short phalanges, extra digits, and in severe cases metacarpal-to-carpal transformation and cortical bone thinning.

Genetic/molecular confirmation. Confirmed by **HOXD13 molecular testing**, including **Sanger/next-generation sequencing** of coding exons plus **triplet-repeat (polyalanine) length analysis** of exon 1 (standard sequencing may miss or misread the GCN repeat). Jin et al. illustrate the approach — STR linkage (D2S1238/D2S1245 flanking HOXD13) plus repeat-length detection of a 27-bp expansion (PMID: 22161087): *"a heterozygous 27 bp expansion within the imperfect GCN triplet-repeat of exon 1, c.184_210dup, was identified."* Whole-exome sequencing (trio-WES) effectively identifies point/truncating variants (PMID: 31611522, PMID: 39639541).

Recommended testing hierarchy: (1) targeted HOXD13 sequencing + repeat-length assay for classic phenotypes; (2) limb-malformation gene panel (HOXD13, GLI3, PTHLH, FBLN1, TP63, LMBR1, GJA1) for atypical/overlapping presentations; (3) WES/WGS for undiagnosed cases; (4) CMA/array-CGH when a contiguous-gene deletion is suspected (e.g., PTHLH-related BDE, PMID: 31283647).

Prenatal diagnosis. Feasible by **combined ultrasound and molecular analysis** (PMID: 22161087): *"Accurate prenatal diagnosis of SPD was achieved with combined ultrasound and molecular analysis."* Amniocentesis-based linkage/mutation analysis at ~18 weeks plus 16–21-week ultrasound of the digits was used successfully.

Omics/other diagnostics. Not routinely used. No specific laboratory biomarkers, metabolites, or circulating markers exist — diagnosis is radiographic and genetic.

Differential diagnosis. Other brachydactylies (BDA3, BDA4, BDE from PTHLH), isolated syndactyly/polydactyly syndromes, acrocephalosyndactyly syndromes (Apert/Pfeiffer — FGFR2, PMID: 40620881, PMID: 42082645), amniotic band sequence (PMID: 41215847), and

syndromic forms with intellectual disability/microcephaly (PMID: 21416592). Distinguishing features: the HOXD13 form is isolated (non-syndromic), affects the autopod with characteristic 4th-web and toe-2/3 involvement, and lacks craniofacial/cognitive/systemic features.

Screening. No population newborn screening (structural, non-metabolic). **Cascade genetic testing** of at-risk family members and **carrier counseling** are appropriate.

Section 11: Outcome / Prognosis

Survival and mortality. Normal life expectancy; no increased mortality. This is a non-lethal structural malformation of the limbs with no systemic organ involvement.

Morbidity and function. Morbidity is limited to **hand/foot function** (grip, dexterity, fine motor skills), **footwear/ambulation** (for foot involvement), and **cosmetic/psychosocial** impact. Disability ranges from negligible (mild brachydactyly) to functionally significant (complex syndactyly/polydactyly requiring reconstruction).

Disease course and complications. Static and lifelong; complications are chiefly those of untreated digital fusion/duplication (impaired dexterity, asymmetric growth) and of surgery (web creep, scarring, need for revision). Early release in selected contexts reduces asymmetric finger growth (PMID: 41215847).

Recovery potential and prognostic factors. Surgical correction generally yields good-to-excellent functional and cosmetic outcomes; e.g., first-web syndactyly series achieved favorable results with dorsal commissural flaps and grafts (PMID: 41030430). Prognostic factors: severity/complexity of the malformation, number of digits involved, presence of osseous (vs cutaneous-only) fusion, and timing/quality of surgery. **Prognostic biomarkers:** none; alanine-repeat length is a genotypic severity correlate rather than a clinical-course biomarker.

Note: One retrospective database study reported an association between **nonsyndromic syndactyly** and increased rates of benign neoplasms and digestive malignancies (PMID: 41797684); this is an epidemiologic signal requiring confirmation and is **not established as a feature of HOXD13-related disease specifically**.

Section 12: Treatment

No disease-modifying pharmacotherapy exists. There is no approved pharmacological, RNA-based, cell, or gene therapy for HOXD13-related limb malformation. Retinoic-acid rescue is an *in utero* mouse finding only (PMID: 19075394) and is **not a human therapy**.

Surgical and interventional (mainstay). Treatment targets the structural anomaly:

- **Syndactyly release** — Z-plasty for simple cases; dorsal commissural/advancement flaps and full-thickness skin grafts for complex/first-web cases (PMID: 41030430): *"Dorsal commissural flaps were the initial operative approach in 100% of the complex/complicated cases, whereas Z-plasties were used in 64% of the simple cases. Full-thickness skin grafts were required in 12 (86%) of the complex/complicated cases."*
- **Polydactyly excision and brachydactyly correction** as indicated.
- Surgical management of syndactyly is well-established (PMID: 40206497): *"Hand syndactyly is a congenital malformation resulting from the failure of differentiation between two or more digits, resulting in a subsequent fusion."*
- **Timing** is individualized; early release is favored in selected contexts to optimize growth and function (PMID: 41215847).

Supportive and rehabilitative. Postoperative **hand therapy / occupational therapy**, splinting/bracing, and dressing care; attention to fine-motor development and psychosocial support.

Experimental / advanced therapeutics. None specific to this disorder. No gene-therapy, ASO, siRNA, or small-molecule programs are in clinical trials for HOXD13 limb malformation.

Suggested NCIT intervention terms: NCIT:C15329 (reconstructive surgery), surgical procedure/hand surgery, NCIT:C15277 (physical therapy), NCIT:C15431 (occupational therapy), NCIT:C15215 (skin graft).

Pharmacogenomics / personalized medicine / combination therapy: not applicable — management is anatomical/reconstructive, individualized to the malformation.

Section 13: Prevention

Primary prevention. Not possible for a spontaneous/inherited Mendelian developmental disorder. **Genetic counseling** is the principal preventive tool: affected individuals have a 50% transmission risk (autosomal dominant), tempered by incomplete penetrance and variable expressivity.

Secondary prevention / screening. Cascade genetic testing of relatives and **prenatal diagnosis** (combined ultrasound + HOXD13 mutation analysis, PMID: 22161087) enable informed reproductive decision-making. **Preimplantation genetic testing (PGT-M)** is applicable where the familial variant is known.

Tertiary prevention. Timely surgical correction and hand therapy to prevent functional disability, asymmetric growth, and contractures.

Immunization, public-health, environmental, prophylactic, behavioral interventions: not applicable — no infectious or environmental component.

Section 14: Other Species / Natural Disease

Model-species orthologs. HOXD13 is deeply conserved. Mouse *Hoxd13* (NCBI Gene ID 15433) is the direct ortholog; chick and zebrafish orthologs exist. Zebrafish HoxD/HoxA cluster genes are expressed during fin development, and comparative studies of digit origin place Hoxd13 at the center of the fin-to-limb transition (PMID: 7791900).

Natural / spontaneous disease in other species. Two **spontaneous mouse mutants** phenocopy human disease: - ***spdh*** (synpolydactyly homolog) — a 21-bp in-frame duplication expanding the polyalanine tract from 15 to 22 residues; homozygotes show polydactyly, syndactyly, and brachydactyly (PMID: 9580668). - ***Dyc*** ("Digit in Y and Carpe") — a trinucleotide expansion (+7 Ala) via FoSTeS, with an SPD-like phenotype (PMID: 19546318).

Comparative biology / evolutionary conservation. The autopod-patterning role of HOXA13/HOXD13 is conserved across tetrapods; the mechanism (homeodomain transcription factor controlling digit morphogenesis) is evolutionarily ancient (PMID: 8898214, PMID: 7791900). **NCBI Taxon:** Homo sapiens (9606), Mus musculus (10090), Gallus gallus (9031), Danio rerio (7955).

Transmission / zoonotic potential: not applicable (non-infectious, genetic).

Section 15: Model Organisms

Mammalian models (mouse — primary).

Model	Type	Lesion	Phenotype recapitulation	PMID
spdh	Spontaneous	PolyAla +7 (21-bp dup)	Polydactyly, syndactyly, brachydactyly; defective chondrocyte differentiation/ proliferation, joint patterning; dominant-negative	9580668, 11850178, 11543619
Dyc	Spontaneous	PolyAla +7 (FoSTeS)	SPD-like limb patterning defect	19546318
Hoxd13 Q50R	Knock-in	Homeodomain missense	Dose-dependent brachydactyly + syndactyly (het) → severe syndactyly/brachydactyly/ polydactyly (homo); SHH/ GREM1/FGF8 up, BMP2 down	38561387
Hoxa13/ Hoxd13 compound	Targeted null	LOF	Severe autopod patterning defects; partial redundancy, dosage control	8898214

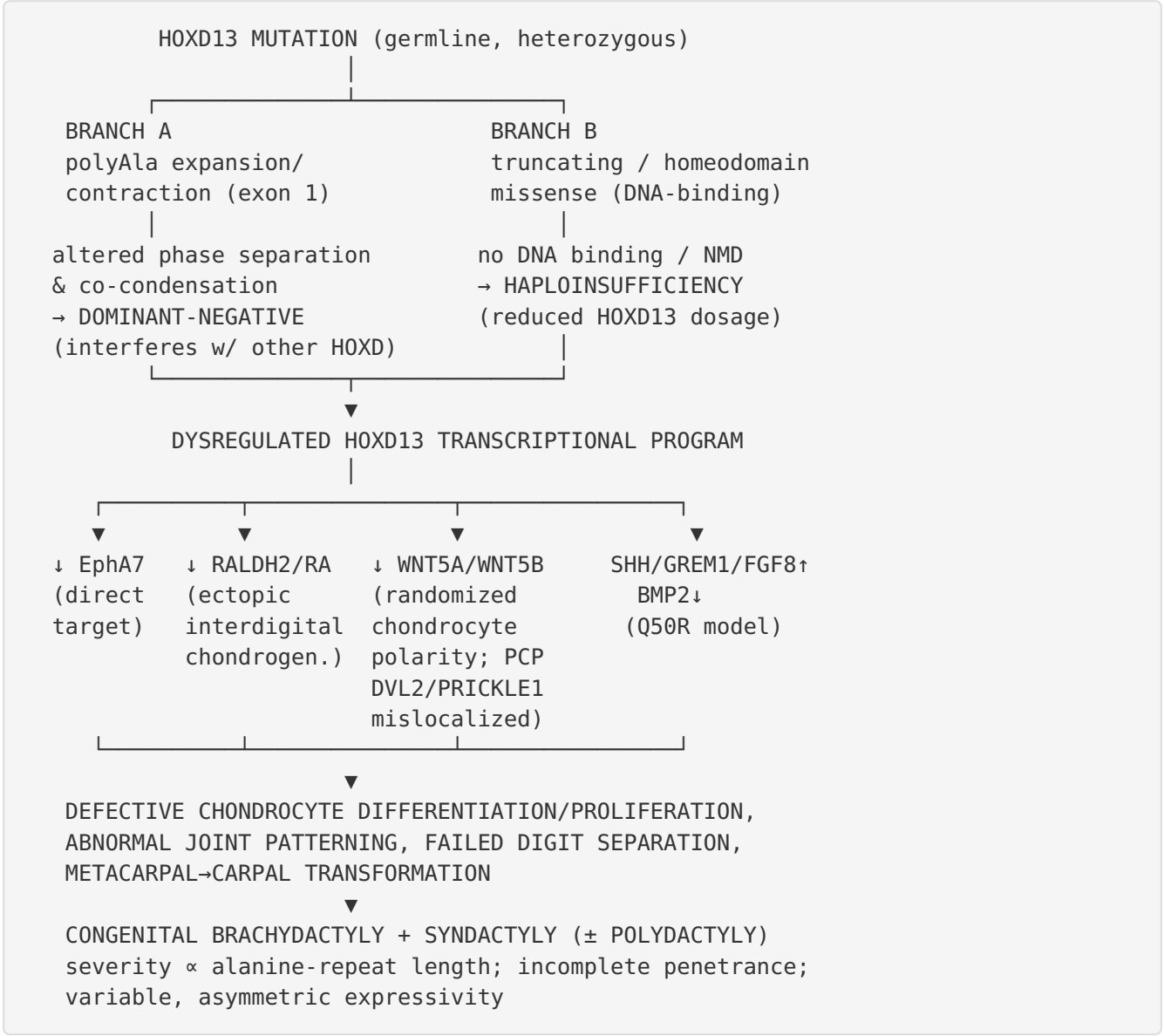
The *Hoxd13* Q50R model faithfully reproduces the human dose-dependent phenotype (PMID: 38561387): *"The mutant mice exhibited obvious limb malformations including slight brachydactyly and partial syndactyly between digits 2-4 in the heterozygotes, and severe syndactyly, brachydactyly and polydactyly in homozygotes."*

Other systems. The **chick embryo** is used for in vivo misexpression studies of HOXD13 variants (e.g., G11A, G11A/GLI3R; PMID: 22373878, PMID: 21782042) and for studying differential cell affinity/sorting in recombinant limb buds (PMID: 12376104). **In vitro** systems include luciferase/EMSA transactivation assays (EphA7, EMSA DNA-binding) and phase-separation/condensate assays (PMID: 32386547).

Model strengths and limitations. Strengths: *spdh* and Q50R reproduce the full brachydactyly-syndactyly-polydactyly spectrum and allow dissection of SHH/GREM1/FGF/BMP2, RA, WNT5A-PCP, and EphA7 mechanisms, plus in utero RA rescue. Limitations: mouse homozygotes are

more severe than typical human heterozygotes; incomplete penetrance and human condensate biology are harder to model; *spdh* males have reproductive-gland anomalies not seen in humans. **Resources:** MGI (Mouse Genome Informatics).

Mechanistic Model / Interpretation



The unifying interpretation is **quantitative homeoprotein dosage control of digit morphogenesis**: HOXD13 output must be precisely tuned, and both too-little functional protein (haploinsufficiency) and a dominant-negative/condensation-altering mutant protein corrupt the same downstream patterning network, producing a graded phenotype.

Evidence Base

PMID	Role	Contribution
17236141	Primary — defining	Defines brachydactyly-syndactyly syndrome; HOXD13 polyAla contraction & Q317R abolishing EPHA7 transactivation
8898214	Foundational	HOXA13/HOXD13 autopod-specific expression; dosage control of digits
20386744	Mechanism	HOXD13–Hand2 complex activates Shh via ZRS
19075394	Mechanism (mouse)	Retinoic-acid reduction → ectopic interdigital chondrogenesis; in utero RA rescue
21814222	Mechanism	Haploinsufficiency for missense/out-of-frame deletions
37427568	Clinical cohort	87 families; penetrance, expressivity, repeat-length/ severity; condensation mechanism
24161848	Mechanism (mouse)	WNT5A-PCP control of chondrocyte polarity; metacarpal→carpal
38561387	Model	Hoxd13-Q50R knock-in; SHH/GREM1/FGF8↑, BMP2↓; dose-dependent phenotype
32789964	Clinical/genetic	Frameshift → BDA3+BDA4
26581570	Clinical/genetic	Homozygous DNA-binding missense; non-penetrant heterozygotes
16314414	Mechanism	EphA7 as direct HOXD13/HOXA13 target; I47L fails to transactivate
22161087	Diagnostics	Prenatal diagnosis via ultrasound + repeat-length analysis
41030430	Treatment	First-web syndactyly surgical techniques/outcomes
40206497	Treatment	Systematic review of hand syndactyly
32386547	Mechanism	Phase-separation/condensate "unblending" for polyAla expansions
9580668, 19546318, 11850178, 11543619	Models/ mechanism	*spdh*/*Dyc* models; dominant-negative; expansion mechanisms

Evidence-type distribution: human clinical-genetic (pedigrees, cohorts), mouse model organism (*spdh*, *Dyc*, Q50R, compound nulls), chick in vivo, and in vitro/biophysical (EMSA, luciferase, condensate assays). The mechanistic claims are strongly convergent across species and methods.

Limitations and Knowledge Gaps

1. **Prevalence/incidence are not precisely quantified** for the specific brachydactyly-syndactyly syndrome; figures are inferred from the broader SPD1 spectrum.
2. **Genotype–phenotype correlation remains imperfect** — incomplete penetrance and asymmetric, variable expressivity (even within families) limit prediction; repeat length correlates with, but does not determine, severity.
3. The **phase-separation/condensate mechanism**, while demonstrated biophysically and in a mouse model, has not been translated into clinical biomarkers or therapeutic targets.
4. **Human therapeutic rescue** of the developmental lesion is not feasible (critical window is embryonic); RA rescue is mouse-only.
5. A reported **syndactyly–neoplasm association** (PMID: 41797684) is from nonsyndromic syndactyly broadly and is **not validated for HOXD13 disease**; it should not be over-interpreted.
6. **Modifier genetics** (what determines penetrance/asymmetry) are essentially unknown.
7. Quality-of-life data are **anecdotal/surgical-series-based**, not systematically measured with validated instruments (EQ-5D, PROMIS).

Proposed Follow-up Experiments / Actions

1. **Natural-history / registry study** of HOXD13-mutation carriers across the allelic spectrum to quantify penetrance, expressivity, asymmetry, and age-specific functional outcomes with validated QoL instruments.
2. **Condensate-targeted mechanistic work** — define how specific repeat lengths quantitatively shift HOXD13 condensate composition and downstream target expression (single-cell/spatial transcriptomics of developing autopod), building on PMID: 32386547.

3. **Modifier mapping** — WGS/GWAS within large pedigrees (and the Gottschalk cohort) to identify genetic modifiers of penetrance and laterality.
4. **Systematic genotype–phenotype database** linking variant class (expansion length, truncation position, homeodomain residue) to standardized HPO-coded phenotypes and surgical complexity.
5. **Functional screens** of VUS in HOXD13 using combined EMSA/EphA7-luciferase and condensate assays to improve ACMG classification.
6. **Surgical outcomes research** — prospective comparison of early vs delayed release and flap techniques for HOXD13-related complex syndactyly with standardized functional/cosmetic endpoints.
7. **Confirm or refute the neoplasm signal** specifically within genetically confirmed HOXD13 cohorts before any clinical surveillance recommendation.

Report compiled from 11 confirmed findings and 44 reviewed papers over a 5-iteration autonomous investigation. All mechanistic and clinical claims are cited to primary literature (PMID). Evidence sources span human clinical-genetic, mouse/chick model-organism, and in vitro/biophysical studies.
